Transfer task: plan and begin

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Organize the given evidence and begin an independent position prediction.

SUCCESS CRITERIA - I can...

1. Organize the task data with fixed axes and matching units.
2. Calculate central and bounded final coordinates.
3. Save an explicit plan for the graph and explanation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “bound” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

bound: an upper or lower limit in a model

assumption: a condition accepted for using a model

01 Warm-up: bring back an earlier idea

A box at rest has 15 N right and 3 N left.
Find the additional horizontal force needed for balance and show the sum.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

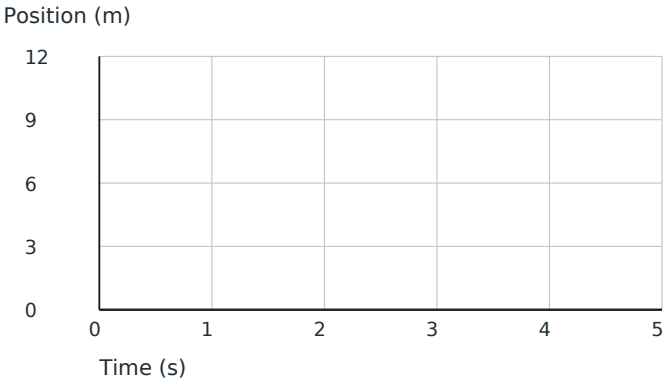
02 Worked example: explain the model

Reference: $x_f = x_i + v_x t$; $y_f = y_i + v_y t$.
Fictional fixed-Earth, constant-velocity model. CPA: x_i = -120,000,000 km; $y_i = 0$; t = 16,000,000 s; v_x = 8.21 km/s; v_y = 2.20 km/s.

Worked reasoning: A simplified model uses x_i=0 m and constant v=2 m/s for 5 s, giving x=10 m. This is a conditional prediction, not proof that the real object follows the model.

Your turn: Draw the predicted path and mark one observation you would collect to test the constant-velocity assumption.

Sentence starter: I predict ... . My model supports this because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Organize the givens

A fictional tracking exercise treats Earth as fixed at the origin. The object moves at constant velocity; gravity and Earth's motion are omitted.
Use the given values to sketch its starting point and both velocity directions.
Label +x toward Earth initially and +y perpendicular to it.

Sentence starter: The … in my model represents … . Its label shows … .

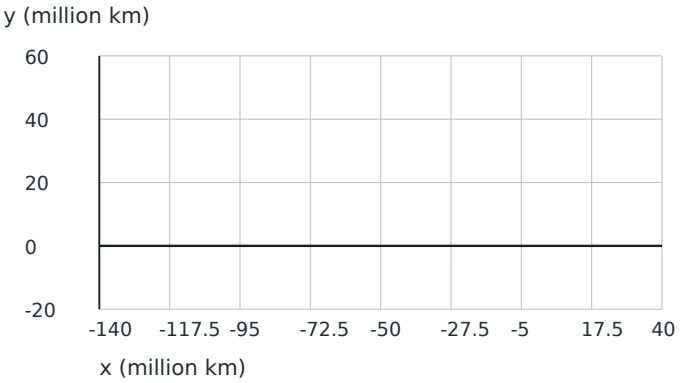
Quantity	Value
Initial x	-120,000,000 km
Initial y	0 km
Time	16,000,000 s
x velocity	+8.21 km/s
y velocity	+2.20 km/s

Marzano check - circle: 1 2 3

04 Find the central end point

Using question 3, find the change in each coordinate and then the final x and y coordinates.
Show each multiplication with units. Plot the starting point and final point on the coordinate grid.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Start the bounds

The same model has independent input bounds: initial x from -120,050,000 to -119,950,000 km; x velocity from 8.02 to 8.40 km/s; y velocity from 2.15 to 2.25 km/s. Initial y and time are exact.
Calculate lower and upper final x and y coordinates.

Sentence starter: I found … [unit]. This value means … in this situation.

Coordinate	Lower (km)	Upper (km)
x	____	____
y	____	____

Marzano check - circle: 1 2 3

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Continue down each column.

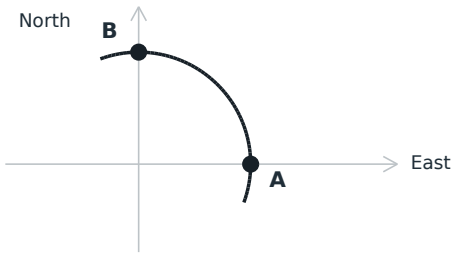
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A rover travels straight east at a steady 7 m/s. A second rover turns at the same steady speed. Which can have zero net force?
Explain using velocity.
Use the curve for the turning case; show how velocity changes.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .



Path sketch: runner moves from A toward B.

Marzano check - circle: 1 2 3

07 Checkpoint

Write a constant-velocity assumption and name one omitted physical effect. List what you finished and the specific next step for Day 22. Save all work.

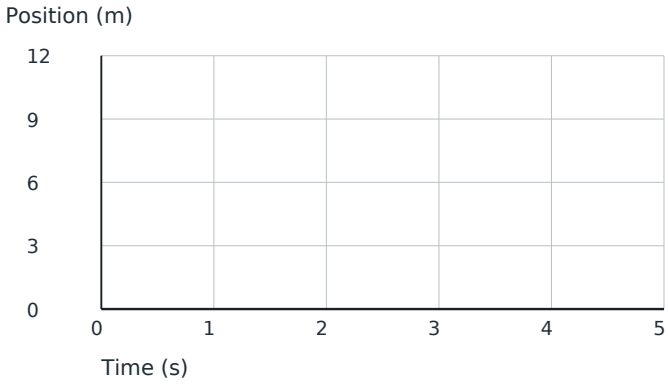
Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Propose two different reasons a measured endpoint could be 8 m.
Give a measurement that would help distinguish them.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3